Databricks

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06026213 BIG DATA SYSTEMS

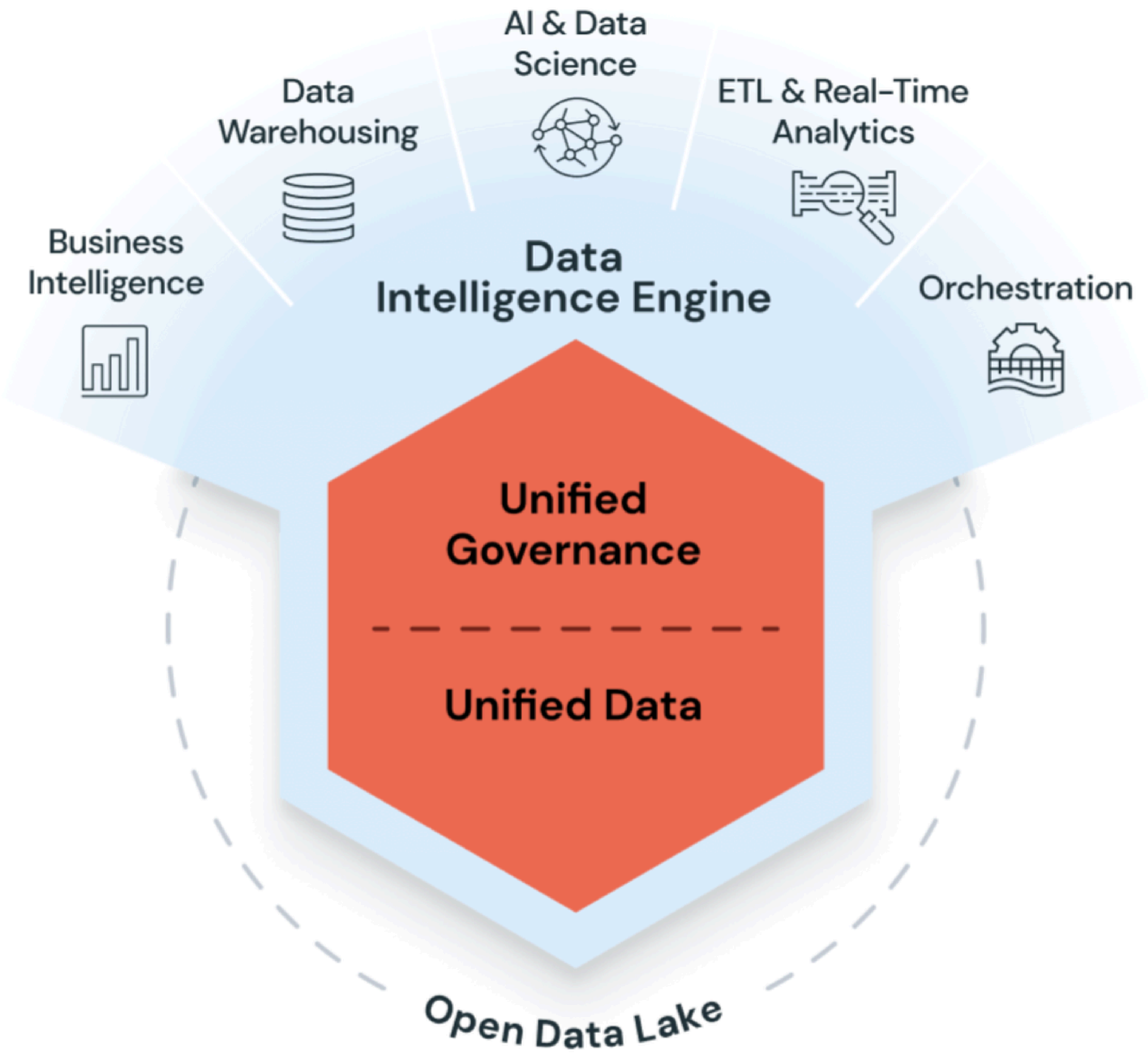
WHAT IS DATABRICKS?

- A **unified**, open analytics platform for building, deploying, sharing, and maintaining enterprise-grade data, analytics, and AI solutions at scale.

- Integrates with cloud storage and security in your cloud account (AWS, Azure, GCP).

- Manages and deploys cloud infrastructure automatically.

databricks

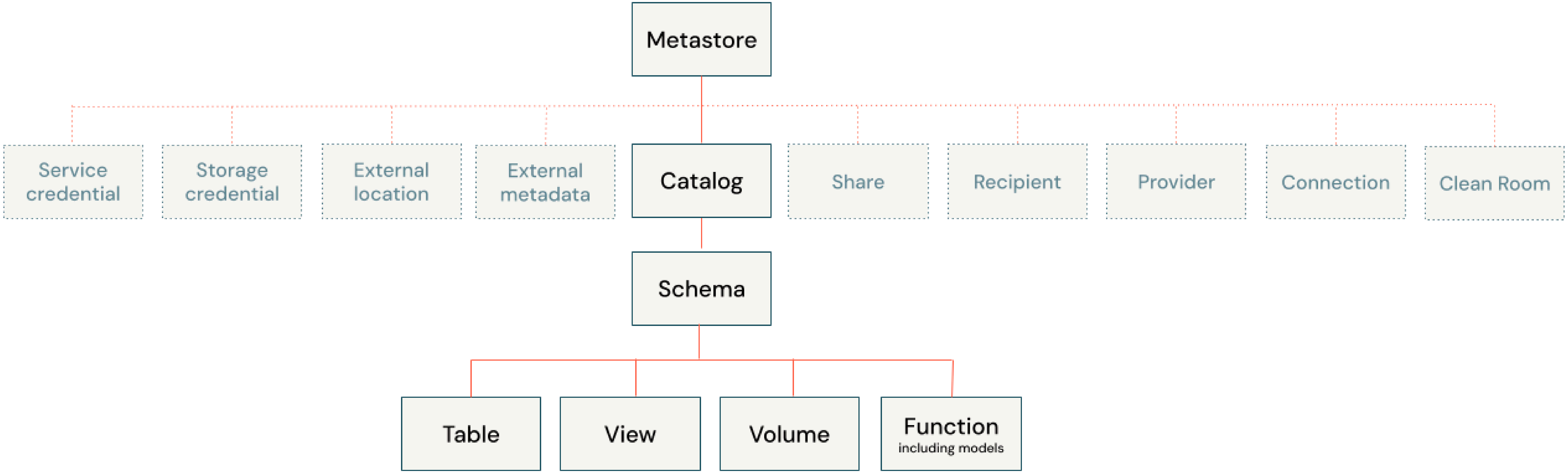


Source: https://docs.databricks.com/aws/en/introduction/

Key Use Cases

- **Enterprise Data Lakehouse**
  - Combines the best of Data Warehouses and Data Lakes.
  - Provides a single source of truth for data engineers, data scientists, and analysts.
  - Eliminates the complexity of maintaining separate systems.
- **ETL & Data Engineering**: Compose logic using SQL, Python, and Scala (Delta Lake integration)
  - Orchestrate job deployment with a few clicks.
- **Machine Learning & AI**: Includes MLflow, runtime for ML, and support for Large Language Models (LLMs).
- **Data Warehousing**: Run analytic queries via SQL warehouses with cost-effective and scalable compute.
- **Data Governance**: Unified governance via **Unity Catalog**.

UNITY CATALOG



“What is the relationship between Apache Spark and Databricks?”

- **Spark** is the core of Databricks platform, powering compute clusters and SQL warehouses.

- **Databricks** provides a managed experience:
    - No need to configure or initialize a Spark context or session manually.

- **Databricks** was founded by the original creators of Apache Spark.

- **Databricks Runtime**: Includes proprietary optimizations and features on top of open-source Spark.
    - **Photon Engine**: An optimized execution layer (written in C++) that vectorizes queries for faster execution than standard Spark.
    - Added optimizations on transformations and actions function

PYSPARK ON DATABRICKS

- The Python API for Apache Spark.
- Allows you to interface with Spark using Python, combining the simplicity of Python with the power of distributed computing.

Key Concept: DataFrames

- DataFrame is a primary object in Spark
  - A DataFrame is a dataset (ex. RDD) organized into named columns.
  - Similar to a table in a relational database or a DataFrame in Pandas
- A **schema** defines the column names and data types

Data Processing

- In distributed systems, **immutability** is key to fault tolerance. Once a DataFrame is created, it cannot be changed—only transformed into a new one.

- **Workflow**: Define a chain of Transformations first, then call a single Action to execute the plan.
    - Lazy Evaluation: Spark optimizes data processing and runs on the cluster